Xiao Liu

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Academic Experience

- **University of Chicago** 2025.8 - Now
  Postdoctoral Scholar, Host: Prof. Chenhao Tan and Prof. James Evans
- **University of California, Los Angeles** 2023.6 - 2023.12
  Visiting Graduate Researcher, Host: Prof. Kai-Wei Chang

Education

- **Peking University** 2020.9 - 2025.6
  Ph.D. in Computer Applied Technology, Advisor: Prof. Yansong Feng
- **Peking University** 2016.9 - 2020.6
  Bachelor of Science in Computer Science (summa cum laude) and Economics

Research Interests

Language model reasoning, Causal reasoning, Trustworthy AI, AI for scientific discovery

Selected Honors

- 2026 Rising Star in Data Science
- 2026 DAAD AInet Fellowship in Reliable AI
- 2026 DSI Faraco Family Postdoctoral Fellowship (1 out of 17)
- 2025 Outstanding Graduate of Peking University
- 2022 Tung Scholarship (1 out of 27)
- 2018 National Scholarship (top 2%)

Publications

* denotes equal contribution; [Google Scholar]

Conference Publications

[c25] **Towards a Universal Causal Reasoner.** NeurIPS 2026 **(Spotlight)**. [PDF] [Data]
Qirun Dai*, **Xiao Liu***, Jiawei Zhang*, Dylan Zhang, Hao Peng, Chenhao Tan

[c24] **Introspection Tools Help LLMs Understand and Control Themselves.** NeurIPS 2026.
Xiao Liu*, Junsol Kim*, Shiyang Lai*, Jacy Reese Anthis*, Siyang Wu, Chenhao Tan, James Evans

[c23] **RefTool: Enhancing Model Reasoning with Reference-Guided Tool Creation.** ICLR 2026. [PDF] [Code]
Xiao Liu, Da Yin, Zirui Wu, Yansong Feng

[c22] **Heuristic-Based Ideation for Guiding LLMs Toward Structured Creativity.** ICLR 2026 Blogpost. [URL]
Xiao Liu*, Haokun Liu*, Chenhao Tan

[c21] **From Single to Societal: Analyzing Persona-Induced Bias in Multi-Agent Interactions.** AAAI 2026 **(Oral)**. [PDF] [Code]
Jiayi Li*, **Xiao Liu***, Yansong Feng

[c20] **HCR-REASONER: Synergizing Large Language Models and Theory for Human-like Causal Reasoning.** AACL 2026. [PDF]
Yanxi Zhang, Xin Cong, Zhong Zhang, **Xiao Liu**, Dongyan Zhao, Yesai Wu

[c19] **Augmenting Research Ideation with Data: An Empirical Investigation in Social Science.** AI4Science Workshop at NeurIPS 2025 **(Spotlight)**. [PDF]
Xiao Liu, Xinyi Dong, Xinyang Gao, Yansong Feng, Xun Pang

[c18] **Recipe2Plan: Evaluating Planning Abilities of LLMs for Efficient and Feasible Multitasking with Time Constraints Between Actions.** EMNLP 2025 (Findings). [PDF]
Zirui Wu, **Xiao Liu**, Jiayi Li, Lingpeng Kong, Yansong Feng

[c17] **Read it in Two Steps: Translating Extremely Low-Resource Languages with Code-Augmented Grammar Books.** ACL 2025. [PDF] [Code]
Chen Zhang*, Jiuheng Lin*, **Xiao Liu**, Zekai Zhang, Yansong Feng

[c16] **QUDSELECT: Selective Decoding for Questions Under Discussion Parsing.** EMNLP 2024. [PDF] [Code]
Ashima Suvarna*, **Xiao Liu***, Tanmay Parekh, Kai-Wei Chang, Nanyun Peng

[c15] **Are LLMs Capable of Data-based Statistical and Causal Reasoning? Benchmarking Advanced Quantitative Reasoning with Data.** ACL 2024 (Findings), AI4Research Workshop 2024 **(Highlight)**. [PDF] [Code]
Xiao Liu, Zirui Wu, Xueqing Wu, Pan Lu, Kai-Wei Chang, Yansong Feng

[c14] **Teaching Large Language Models an Unseen Language on the Fly.** ACL 2024 (Findings). [PDF] [Code]
Chen Zhang, **Xiao Liu**, Jiuheng Lin, Yansong Feng

[c13] **CASA: Causality-driven Argument Sufficiency Assessment.** NAACL 2024. [PDF] [Code]
Xiao Liu, Yansong Feng, Kai-Wei Chang

[c12] **Dynosaur: A Dynamic Growth Paradigm for Instruction-Tuning Data Curation.** EMNLP 2023. [PDF] [Code]
Da Yin*, **Xiao Liu***, Fan Yin*, Ming Zhong*, Hritik Bansal, Jiawei Han, Kai-Wei Chang

[c11] **DiNeR: A Large Realistic Dataset for Evaluating Compositional Generalization.** EMNLP 2023. [PDF] [Code]
Chengang Hu, **Xiao Liu**, Yansong Feng

[c10] **The Magic of IF: Investigating Causal Reasoning Abilities in Large Language Models of Code.** ACL 2023 (Findings). [PDF] [Code]
Xiao Liu, Da Yin, Chen Zhang, Yansong Feng, Dongyan Zhao

[c9] **How Many Answers Should I Give? An Empirical Study of Multi-Answer Reading Comprehension.** ACL 2023 (Findings). [PDF] [Code]
Chen Zhang, Jiuheng Lin, **Xiao Liu**, Yuxuan Lai, Yansong Feng, Dongyan Zhao

[c8] **Counterfactual Recipe Generation: Exploring Compositional Generalization in a Realistic Scenario.** EMNLP 2022. [PDF] [Code]
Xiao Liu, Yansong Feng, Jizhi Tang, Chengang Hu, Dongyan Zhao

[c7] **Things not Written in Text: Exploring Spatial Commonsense from Visual Signals.** ACL 2022. [PDF] [Code]
Xiao Liu, Da Yin, Yansong Feng, Dongyan Zhao

[c6] **Dual-Channel Evidence Fusion for Fact Verification over Texts and Tables.** NAACL 2022. [PDF] [Code]
Nan Hu, Zirui Wu, Yuxuan Lai, **Xiao Liu**, Yansong Feng

[c5] **Everything Has a Cause: Leveraging Causal Inference in Legal Text Analysis.** NAACL 2021 **(Oral)**. [PDF] [Code]
Xiao Liu*, Da Yin*, Yansong Feng, Yuting Wu, Dongyan Zhao

[c4] **Neighborhood Matching Network for Entity Alignment.** ACL 2020 **(Oral)**. [PDF] [Code]
Yuting Wu, **Xiao Liu**, Yansong Feng, Zheng Wang, Dongyan Zhao

[c3] **Interactive Multi-grained Joint Model for Targeted Sentiment Analysis.** CIKM 2019 **(Oral)**. [PDF]
Da Yin*, **Xiao Liu***, Xiaojun Wan

[c2] **Jointly Learning Entity and Relation Representations for Entity Alignment.** EMNLP 2019. [PDF] [Code]
Yuting Wu, **Xiao Liu**, Yansong Feng, Zheng Wang, Dongyan Zhao

[c1] **Relation-aware Entity Alignment for Heterogeneous Knowledge Graphs.** IJCAI 2019 **(Oral)**. [PDF] [Code]
Yuting Wu, **Xiao Liu**, Yansong Feng, Zheng Wang, Rui Yan, Dongyan Zhao

Journal Publications

[j2] **Eliciting and Improving the Causal Reasoning Abilities of Large Language Models with Conditional Statements.** Computational Linguistics 2025. [PDF]
Xiao Liu, Da Yin, Chen Zhang, Yansong Feng, Dongyan Zhao

[j1] **A Graph Learning Based Approach for Identity Inference in Dapp Platform Blockchain.** IEEE Transactions on Emerging Topics in Computing 2020. [PDF]
Xiao Liu, Zaiyang Tang, Peng Li, Song Guo, Xuepeng Fan, Jinbo Zhang

Working Papers

[w4] **Contemporary AI lacks the imagination to diverge or negate in science.** Nature R&R. [PDF]
Honglin Bao, Siyang Wu, **Xiao Liu**, Sida Li, Shiyun Cao, James Evans

[w3] **Building an Atlas of Social Experiments to Link Studies, Reconcile Conflicts, and Bridge Gaps.** [PDF]
Jiawei Zhang*, Honglin Bao*, Pengda Wang*, Alex Yan*, **Xiao Liu**✉, James Evans

[w2] **Markets, Not Planners: Decentralized Orchestration of LLM Agents with Private Information.** [PDF]
Xiao Liu*, Haoyang Li*, Songwei Li, Hongbo Fang, Fengli Xu, Feng Shi, James Evans

[w1] **CausaLab: A Scalable Environment for Interactive Causal Discovery Toward AI Scientists.** [PDF]
Junlin Yang, Dylan Zhang, Xiangchen Song, Qirun Dai, **Xiao Liu**, Yuen Chen, Aniket Vashishtha, Jing Shi, Chenhao Tan, Hao Peng

Teaching

- **Data Science Clinic** — Winter 2026, Spring 2026
  Mentor of the project with Accountability Counsel, featured in [News] — *University of Chicago*
- **Data Thinking and Analytical Methods for the Social Sciences** — Spring 2025
  Workshop instructor and textbook co-author — *Peking University*
- **Foundations of Natural Language Processing** — Spring 2024
  Teaching Assistant — *Peking University*
- **Empirical Methods in Natural Language Processing** — Spring 2023
  Teaching Assistant — *Peking University*

- **Introduction to Computation** Fall 2020
  Teaching Assistant *Peking University*

Service

- **Co-Organizer** AI & Scientific Discovery Online Seminar
- **Area Chair** ACL Rolling Review 2024-2026
- **Senior Program Committee** AAAI 2026
- **Reviewer** ICLR, NeurIPS, ACL, EMNLP, NAACL, COLM, CSCW, Computational Linguistics, TKDD, TKDE, IPM
- **Volunteer** ACL, EMNLP, NAACL